Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

DEPARTAMENTO DE FÍSICA

TRABAJO FIN DE GRADO

Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CEνNS

Realizado por
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Para la obtención del título de

Maestra en Ciencias
en la Especialidad de Física

Dirigido por

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Ciudad de México, Noviembre de 2026

Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

PHYSICS DEPARTMENT

Sensitivities to neutrino charge radius at reactor neutrino experiments involving CEνNS

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In order to obtain the

Master of Science degree
speciality in Physics

Under the supervision of

PhD Omar Gustavo Miranda Romagnoli

Mexico City, November of 2026

Agradecimientos

agradecimiento aqui.

Resumen

resumen aqui

Abstract

resumen en ingles

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1. Introduction

intro

2. Electroweak theory in the Standard Model of particles

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter particles and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^{\pm}$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e, ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^-, μ^- and τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

2.2.1 Neutrinos

Neutrinos are charge-less, color-less, spin-$\frac{1}{2}$ particles with negligible masses compared to other fundamental particles first proposed in 1930 by Pauli W. [1], and later detected by C. L. Cowan et al. in 1956 [2], to solve known apparent contradictions in conserved quantities of beta decays. Beta decay was known to be as the emission of beta particles (electrons) by the nuclei of unstable atoms, wherein the difference in mass of the original and resulting nuclei of the decay should fix the energy at which the emitted particles are detected. Also of note, due to conservation of angular momentum the sum of the spin (S_e) of the electron, the spin of the resulting nucleus (S_{N^*}) and the angular momentum between them (L) should equal the original nucleus' spin (S_N). Despite this, it was know experimentally that the resulting electrons where emitted in a spectrum of energies less than expected and at $L = 0$, making the sum of spins impossible to match. Pauli's solution to the apparent violation of the conservation of these quantities was to hypothesize the existence of a second neutral spin-$\frac{1}{2}$ with very small mass compared to the electron that allows for both the distribution of energy between both emitted particle, thus letting the electron be emitted at different energies depending on the kinematics, and the conservation of

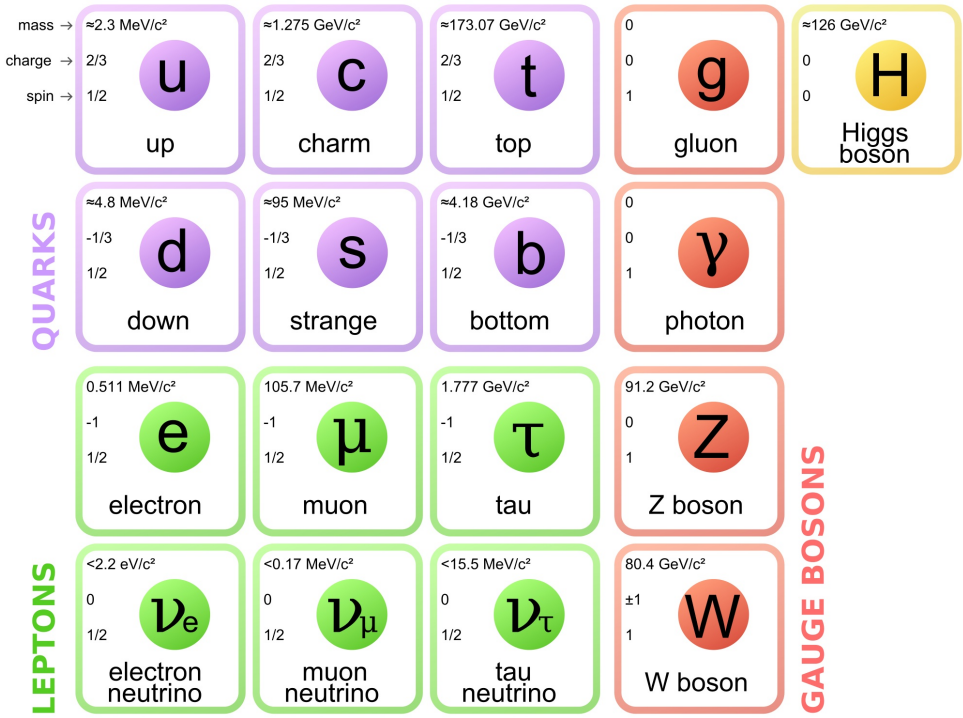


Figure 2.1: *Particles in the SM*

angular momentum.

$N \rightarrow N^* + e^-$	$\rightarrow$	$N \rightarrow N^* + e^- + \nu$
$m_N = m_{N^*} + E_{e^-}$	$\rightarrow$	$m_N = m_{N^*} + E_{e^-} + E_\nu$
$S_{z\ N} = S_{z\ N^*} + S_{z\ e^-}$	$\rightarrow$	$S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} + S_{z\ \nu}$

Later refinement of the atomic model found that nuclei where made up of positively charged elements named protons (p) and neutrally charged elements known as neutrons[1] (n), leaving the small hypothesized particle to be named neutrino (ν) later by Enrico Fermi. It is well understood now that the beta decay is a process in which a neutron within an unstable nucleus decays into a proton, an electron and a(n) (anti)neutrino.

$$n \rightarrow p + e^- + \bar{\nu}$$

Subsequent experiments exploiting cross symmetry of the above interaction [2], accelerator-produced pion decay [3], and charged-current tau neutrino interactions in nuclear emulsion [4, 5] confirmed the existence of three distinct active neutrino flavors and the conservation of lepton family number in the interactions that involved these particles. Given that neutrinos don't interact via electromagnetic or strong forces a theory of a different force that mediated these interactions was developed into what is now known as the "weak nuclear force".

2.2.2 Weak Nuclear Force

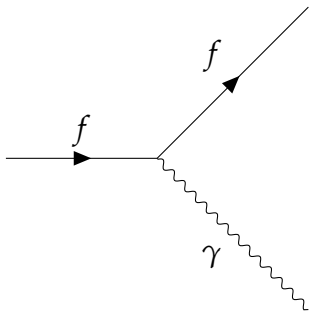
Developments in quantum mechanics (QM) and special relativity (SR) led up to the first theories that could describe the behavior of some subatomic particles and their

[1] this was the name first proposed for the particle hypothesized by Pauli, later changed

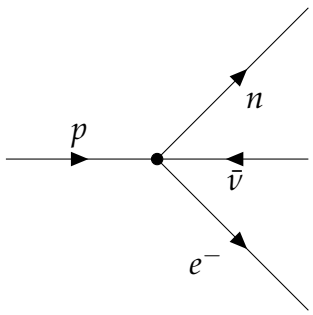
interactions through excitations of quantized fields called quantum field theories (QFTs) obtained by promoting classical fields to operator-valued quantities satisfying canonical (anti)commutation relations and deriving their dynamics from a Lorentz-invariant action principle. One of the earliest and most successful QFTs was Quantum Electro Dynamics (QED), describing how spin-1/2 (anti)particles ($\overset{(-)}{\psi}$) interacted with an EM 4-potential (A^μ) by minimizing the action given by the lagrangian density:

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi}_{\text{free Dirac field}} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free EM field}} - \underbrace{e\bar{\psi}\gamma^\mu\psi A_\mu}_{\text{interaction}}. \tag{2.1}$$

While the 'free' terms in (2.1) describe the propagation of Dirac fermions and photons in free Minkowski space, the 'interaction' term describes a vertex that couples the (anti)fermion current with the electromagnetic four-potential, representing the elementary process of a(n) (anti)fermion emitting or absorbing a photon. At zeroth-order $\hbar$ perturbative terms, QED had shown predictive power for processes like Compton ($\gamma + e^- \to \gamma + e^-$) [6] and Møller ($e^- + e^- \to e^- + e^-$) [7] scattering by the time Fermi E. first proposed an interaction inspired by the QED interaction term, where instead of a fermion current emits a photon, two fermion currents meet at a vertex such that a coupling constant (now called Fermi's constant G_F) quantified the intensity of the interaction, and thus can be estimated experimentally.



$$\sim \quad e\bar{\psi}_f\gamma^\mu\psi_f A_\mu$$



$$\sim \quad \frac{G_F}{\sqrt{2}}(\bar{\psi}_n\gamma^\mu\psi_p)(\bar{\psi}_\nu\gamma_\mu\psi_{e^-})$$

2.3. Massive and Massless Mediators

2.4. Electroweak mixing $\sin^2\theta_W$

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5.1. Results

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